Minos2 – A Guide to Setting up RigControl

Introduction

Welcome to the wonderful world of trying to make your Computer control your Rig.

I'm Peter Burton G3ZPB, and a long-term user of computers in Amateur Radio. I'm a recently-joined member of the Minos Version 2 Development Team. My contribution is limited to testing some of the "Beta" version releases and assisting with the Documentation, Help Files and authoring this document.

In this document, I will attempt to cover some of the problems you might come across while configuring the "RigControl" section of Minos 2. I strongly suggest you read through the whole document first, before setting up RigControl. Many of the points will also apply to "RotatorControl".

It is specifically written for Minos v2.0.8.Beta running on a computer with the MS Windows 10 operating system. It will be updated as newer versions become available.

First and foremost you should read, understand and follow the procedure set out in the excellent "*Documentation for Minos Version 2 Contest Logging Software*" written by Ken Punshon, G4APJ.

The second thing is **don't panic** if it doesn't all work first time, within 5 minutes - unfortunately, the setting up of COM ports so they work properly on MS Windows computers seems to be an Art not a Science!

References

1> Minos software at https://sourceforge.net/projects/minos/files/
2> Documentation at https://sourceforge.net/projects/minos/files/
3> Typical RigControl settings at https://minos.groups.io/g/users/files
4> Typical RotatorControl settings at https://minos.groups.io/g/users/files

Installation

This installation procedure is for a "first-time" installation assuming no Rig (or Contest) has been setup previously. You must follow this sequence in order to have a reasonable chance of success at the first attempt.

Connect the Rig to your computer via a suitable interface (USB or Serial or Hybrid e.g. USB to Serial).

Switch on the computer and when the all the "Start-up" programs have loaded, switch on the Rig.

The following text is based on Ken Punshon's document reference <2> above.

Installation of RigControl should be carried out as follows…

Install the Minos2 software as detailed in Chapter 1 (if not already done so.)

1> Check the settings of your rig and note the following parameters:
   a. COM port number e.g. COM6
   b. * Serial Speed e.g. 19200Bd
   c. Number of Data Bits e.g. 8
   d. Number of Stop Bits e.g. 1
   e. Parity e.g. None

f. * Handshake e.g. CTS/RTS

g. * For Icom rigs only, CIV setting e.g. 0x7c

h. These settings are usually found buried in a "Menu Settings" of the Rig somewhere (you may need to read the manual to find where). Don't take for granted that the settings will be the "Default values" shown in the manual, even for a new rig. The parameters indicated * are nearly always resettable from within the Rig, the others may not be.

2> Setup Minos Control Applications as follows:

a. Start-up Minos Logger (if not already running) and select OK from the Welcome screen

b. From the main screen go to Tools/Start-up Apps Configuration

c. Click on New Entry; Select App Type = Server (you must have "Server" running for any of the Additional Apps to operate)

d. Click on New Entry; Select App Type = RigControl

e. Make a note of the "Server Name" (and change if required)

f. Click on Save and Start Apps

3> Setup RigControl:

a. Open the Configure/Setup Radios option in Minos Rig Control

b. Click on Add Radio

c. Enter a name for the Radio to be controlled and note it

d. Select the Radio Model from the pull down list (this list is provided by "HamLib" and covers most models likely to be used; new models are added as they become available). Select OK

e. Select Comport, Speed, Data Bits, Stop Bits, Parity, Handshake etc. (and CIV if applicable) as noted in step <1b to 1g> above. Select OK

f. Select Cancel in Minos App Configuration screen

4> Setup a Test Contest for trial purposes:

a. Click on File; Select New Contest in main Minos Contest Logger window

b. Type in "Test" for Contest Name and select the appropriate Band

c. Type something suitable for Callsign as sent, Locator as sent and Power

d. In Radio Name, select the previously setup Radio from the pull-down list. Select OK

e. Save the new contest log file

f. After a brief pause, the Rig frequency should be correctly displayed in the RigControl window.

If it doesn't work

If the settings appear to be correct and the Minos programme refuses to communicate with your Rig, switch everything off, go and get a cup of coffee and enjoy it. When finished, come back, switch on the computer and when the all the "Start-up" programs have loaded, switch on the Rig. Then start the Minos programme; hopefully it will all be working for you. Seriously! Some settings seem not to take effect until a complete computer restart is carried out.

If it still doesn't work, it is hoped the following points may assist you…..

1> We have found that using direct USB connection between computer and rig is likely to be more reliable that using an RS232 connection.

2> Using an un-powered USB2 Hub seems to offer an improvement in some cases. For example, connect the Rig (and Rotator) and Mouse to a USB Hub and then connect the Hub to one port on the computer.

3> If using a USB<->RS232 converter, ones using a genuine "Prolific" chipset seem to be more reliable than others.

4> Ensure your Rig is set to the same Data Speed (Baud Rate) as the programme! On some Rigs, you need to switch off the power and back on again for the change to take effect. Most Rigs will work over a wide range of Baud Rates (provided the Rig and Computer are set to the same speed!) It is better to start with a slow speed e.g. 4800Bd. When everything is working you can them try moving up to a faster speed e.g. 38400Bd for a more “responsive” system.

5> Values for some of the other parameters are more variable, specifically:
CI-V (Icom only) – this can usually be left blank as the default value is almost invariably used.
Handshake – try starting with “None”. If this doesn’t work, try “CTS/RTS” next.

6> If you are using other software that controls your rig (e.g. WSJT-X), check the settings that work there and try them with RigControl.

7> Problems can start again after other software has been installed, particularly if new "Virtual Port Drivers" are installed.

8> Rebooting the computer can sometimes help solve a problem.

9> Sometimes rebooting is not enough, so switching off the Computer (and Rig) then powering up again may solve the problem.

10> There are sometimes problems with “sharing” and other conflicts; it is worthwhile disconnecting everything else connected to your computer except Monitor, Keyboard, Mouse and the Rig. Also close down any other programmes you have running. You can always reconnect/restart others after you have successfully got Minos running. Specifically, we have found problems on some computers that will not allow HamLib and OmniRig to run together.

If/when you have solved a Comms problem, would you please email me the details so I can add them to the RigControl list which I will then publish at intervals for others to use.

Email to MinosRigData@outlook.com

73, Peter G3ZPB.

Annex

I strongly suggest you write down the settings for your rig here…..

Parameter	Value	Example
Server name		Dell-E6430
Radio name		IC9000
HamLib Model number		368
CI-V (Icom only)		0x7c
COM port		6
Speed		19200
Data bits		8
Stop bits		1
Parity		None
Handshake		CTS/RTS